

Artificial Intelligence Auditability and Auditor Readiness for Auditing Artificial Intelligence Systems

Abstract

As the business community races to implement artificial intelligence (AI), there are several challenges that need to be addressed such as fairness and biases, transparency, denial of individual rights, and dilution of privacy. AI audits are expected to ensure that AI systems function lawfully, robustly, and follow ethical standards (e.g., fairness). While the auditability for financial audits and information system audits has been well addressed in the literature, auditability of AI systems has not been sufficiently addressed. AI auditability and auditors' competencies are crucial for ensuring AI audits are conducted with high quality. Research on the auditability of AI and the competencies of AI auditors is gravely lacking leaving risks in AI systems unmitigated. The primary reason is that the field is nascent and the rapid growth has left the audit profession struggling to catch up. Foundational work on establishing parameters for such research would help advance this research. In this paper, we explore AI auditability measures and competencies required for conducting AI audits. We conducted semi-structured interviews with 23 experienced AI professionals who have direct involvement or indirect exposure to AI audits. Based on our findings, we propose a framework of AI auditability and identify the competencies required to conduct AI audits. Our study serves as the first formal attempt to systematically identify and classify auditability measures and auditors' expertise demanded for AI audits based on practitioners' perspectives. Our findings contribute to the AI audit literature, inform AI developers about implementing auditability, guide the training of new AI auditors, and establish a foundation for further research in the field.

Keywords: AI audits, auditability, auditors, competency, AI auditability framework

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1. INTRODUCTION

AI systems have gained significant traction in the market over the last decade. However, widely deployed AI systems have caused human rights issues and increased risk of discrimination against minority groups. One example is the racial discrimination in the Correctional Offender Management Profiling for Alternative Sanctions system applied in the U.S. court system (Corbett-Davies, Pierson, Feller, & Goel, 2016). Beyond fairness issues, there are other potential harms caused by AI systems, such as non-transparent, unexplainable, or unjustifiable outcomes, invasions of privacy, etc. (Ayling & Chapman, 2022; Leslie et al., 2019). To address potential deleterious impacts, AI systems need to be audited to ensure lawfulness, technical robustness, and ethical design and use. Regulatory authorities and enterprises have initiated the design of AI audit frameworks and have attempted to define AI principles that guide audit activities (e.g., European Union (EU), 2022; Information Commissioner's Office (ICO), 2020; Personal Data Protection Commission (Singapore) (PDPC), 2020; Raji et al., 2020; Supreme Audit Institutions (SAI), 2020; The Institute of Internal Auditors (IIA), 2017). Auditing AI systems against a set of laws and industry standards ensures AI lawfulness (PricewaterhouseCoopers (PwC), 2019), whereas ethics-based AI auditing informs individuals why a decision was reached and how to contest it, allowing for AI governance, anticipating and mitigating harms, allocating accountability, and balancing conflicts of interest (Mökander & Floridi, 2021). Big Four accounting firms have developed frameworks to help clients design and deploy trustworthy AI and manage AI risks (e.g., Deloitte, 2020; PwC, 2019). Hence, how to conduct an effective and efficient audit that ensures AI ethics, lawfulness, and technical robustness is of great interest for IS researchers and practitioners.

In the information system (IS) literature, efforts have been observed in exploring responsible AI (Dennehy et al., 2023; Dwivedi et al., 2021; Stahl, 2022) and understanding how

to mitigate negative outcomes associated with AI implementation (Mikalef, Conboy, Lundström, & Popovič, 2022). Common ethical concerns about AI used in accounting systems are associated with data security, accountability, and accessibility, model transparency, accountability, and trustworthiness (Zhang et al., 2023), which may trigger audits for AI-powered accounting information system ethics. As AI solutions are increasingly applied to enhance the efficiency of accounting information systems (Qasaimeh, Yousef, Al-Gasaymeh, & Alnaimi, 2022), remove internal control systems weakness (Askary, Abu-Ghazaleh, & Tahat, 2018), predict management accounting information (Chowdhury, 2023), etc., auditing AI-powered accounting information systems may demand additional measures that facilitate the audits and additional auditor expertises. However, current literature on improving audit effectiveness and efficiency for AI systems, including AI-powered accounting information systems, is greatly lacking.

Conducting a highly effective and efficient AI audit that ensure ethical, lawful, and technically robust AI systems demand auditable AI systems and competent auditors. Auditability, which refers to a set of principles that enable audit procedures to be performed, is essential to the widespread development and deployment of AI systems (Liu et al., 2021). Auditability has been identified as one of the major quality drivers of traditional financial and Information Technology (IT) audits; much previous information system research has urged information system scholars to focus more on designing AI auditability (Liang et al., 2021; Robert et al., 2020). However, auditability, as with all other high-level AI principles, is subject to difficulty regarding interpretation and translation into actionable measures. The current literature lacks systematic and practitioner-verified auditability measures to be implemented over the lifecycle of AI systems. In addition to having an auditable AI system, an audit team with diverse expertise is also required for a successful AI audit. Existing research suggests that auditing an AI system involves a

collaborative effort between IT auditors, data scientists, product managers, data engineers, domain experts, and delivery managers (Leslie et al., 2019; SAI, 2020). This assertion seems ad hoc and lacks practical evidence, and does not clarify the roles of these individuals on the audit team. Additionally, given the extensive AI audit objectives expected by regulators, industries, and organizational and individual users, this list of necessary expertise seems incomplete.

In this study, we address gaps in the literature by comprehensively establishing 1) auditability measures for AI systems and 2) the competencies required for audit teams to conduct AI audits based on auditors' practical experiences. The scope of this study involves audits that are being conducted to ensure AI lawfulness, technical robustness, and ethics. Using semi-structured interviews with 23 AI audit experts who had exposure to and knowledge of AI audits, the objective of the current study is to: 1) understand the current state of AI audit practices; 2) establish auditability measures for AI systems based on an auditors' perspective; and 3) identify the competencies required for conducting AI audits during the audit process. We develop an auditability framework that will guide the future design of AI systems to have auditability built in the design, development, deployment, and use phases. This study also informs the AI audit profession and universities on how to train future generations of AI audit professionals.

In the following text, we first present a literature review on 1) AI regulations, 2) AI auditability research, and 3) the competencies required for AI audits. Then, we develop the methodology, including the data collection, interview protocol, and participants. Third, we report our qualitative findings based on 23 semi-structured interviews. Fourth, we discuss our findings and implications for research and practice. Finally, we provide a conclusion.

2. LITERATURE REVIEW

2.1 AI Regulations and AI Audit Frameworks

Over the last few years, regulatory efforts in the AI field have been observed in the U.S. and Europe. In the U.S., the National Institute of Standards and Technology (NIST) is developing the *AI Risk Management Framework (AI RMF)* intended for the voluntary improvement of AI trustworthiness in the AI design, development, use, and evaluation phases (NIST, 2022). IIA's *Artificial Intelligence Auditing Framework* focuses on AI strategy, governance, and human factors, it is intended to be used for internal auditors' advisory, assurance, or blended advisory and assurance services (IIA, 2017). U.S. Executive Order (EO) 13960 establishes principles for the use of AI in the federal government in the United States (Executive Office of the President, 2020). The Committee of Sponsoring Organizations of the Treadway Commission (COSO) and Deloitte co-authored a resource to assist organizations in designing and implementing governance, risk management, oversight strategies, and structures associated with AI (Calagna, Cassidy, & Park, 2021).

The *AI Act* was the first law on AI proposed by a major regulator (i.e., EU) worldwide. In this effort, AI systems are divided into unacceptable-risk, high-risk, and non-high-risk categories; the *AI Act* aims to ensure that high-risk AI systems placed and used on the EU market are lawful, safe, and trustworthy (EU, 2022). As of now, formal law has not yet been enacted. Other regulatory efforts and guidelines developed in Europe also target AI systems, such as ICO's "Guidance on the AI auditing framework" (ICO, 2020), the High-Level Expert Group on Artificial Intelligence (AI HLEG)'s "Ethics Guidance for Trustworthy" (Cannarsa, 2021), and the European Committee for Electrotechnical Standardization (CENELEC)'s "foresight on standardization" exercises that address data quality and related ethical concerns (Balahur et al., 2022).

Standards for AI are also being developed in other countries and internationally. For example, the government of Canada published guiding principles in its *Responsible Artificial*

Intelligence (version 2.0) to ensure the effective and ethical use of AI in 2018 (Government of Canada, 2018). Singapore's government released a *Model Artificial Intelligence Governance Framework (2nd Edition)* to regulate AI systems (PDPC, 2020). In China, the Standards Administration of China released a *White Paper on AI Standardization* that emphasizes the safety, ethical, and privacy issues of AI; therefore, it calls for supervisory control and audit activities to be carried out to ensure that the system meets policy, legal, resource, and business requirements (China Institute for Science and Technology Policy at Tsinghua University, 2018). The Organization for Economic Co-operation and Development (OECD) published recommendations for AI that include trustworthy AI principles and called for AI trustworthiness to be promoted through national policies and international co-operation (OECD, 2019).

Table 1 provides a summary of the discussed regulations. Currently, many regulations emphasize AI accountability, transparency, fairness, explainability, safety, data security and privacy. However, fewer regulations have addressed data quality and traceability issues which have also become critical today. While U.S. regulations and standards related to AI has made explainability a core principle, European regulations and standards do not focus much attention on AI explainability. As AI regulations are emerging, there is a lack of consensus on high-level ethics principles. There are imbalances in AI regulations across countries and industries, leading to confusion in establishing the scope of AI audits.

AI ethics, responsibility, and trustworthiness are used interchangeably in the current literature to address the ethical principles that AI should follow (Berghoff et al., 2022; Dennehy et al., 2023). While AI lawfulness and technical robustness are concerns, there is an emerging stream of research that focuses on ethics-based auditing, since Brundage et al. (2020) suggested that third-party auditors can provide assessments for the safety, security, privacy, and fairness of AI. The

goal of ethics auditing of AI is to improve the ethical performance and meaningful accountability of the audited organization (Schöppl, Taddeo, & Floridi, 2022) and help bridge the gap between principles and practices in AI ethics (Mökander & Floridi, 2022). Shneiderman (2020) sees the assurance of AI trustworthiness as a third layer of governance that is conducted by external review organizations in addition to AI reliability and safety layers, which are assured by organizations internally. According to Shneiderman (2020), the trustworthiness of AI is enhanced and monitored by government interventions and regulations, accounting firms conducting external audits, insurance companies compensating for failures, non-governmental and civil society organizations advancing design principles, and professional organizations and research institutes developing standards, policies, and novel ideas.

In this study, following the current regulatory landscape for AI systems, we explore the auditability measures and auditors' competencies for auditing AI ethics (i.e., accountability, transparency, fairness, safety and security, data privacy and governance, explainability), AI lawfulness, and technical robustness.

Table 1: A Summary of AI Principles according to Different Regulations

Regulations	Responsibility, Ethics, and Accountability	Transparency and Traceability	Fairness, Bias Identification, and Mitigation	Safety and Security	Data Privacy and Data Governance	Validity, Reliability, Quality, and Robustness	Lawfulness	Explainability and Interpretability
The AI Act	Accountability	Transparency	Non-discrimination diversity; fairness	Security; resilience; safety	Data privacy	Technical robustness		
ICO Guidance on AI auditing framework	Accountability	Transparency in data processing	Fairness	Security			Lawfulness; respect for individual rights	
AI HLEG's Ethics Guidance for Trustworthy AI		Explicability	Fairness	Harm prevention			Respect for human autonomy	
CEN and CENELEC	Ethics				Data quality			
IIA's AI Auditing Framework	Ethics			Cyber resilience	Data quality	Performance		Black box elements
Responsible AI in the Government of Canada	Ethics; responsibility	Transparency	Fairness	Data security	Data privacy		Lawfulness	Explainability
Model AI Governance Framework (Singapore)		Transparency	Fairness				Human-centric nature	Explainability
China's White Paper on AI Standardization	Ethics			Safety	Privacy			
NIST AI RMF	Accountability	Transparency	Fairness; bias is managed	Security; resilience; safety	Privacy-enhanced	Validity; reliability		Explainability; interpretability
OECD AI Recommendation	Accountability; responsible disclosure	Transparency	Fairness	Security; safety		Robustness	Human values	Explainability
EO 13960	Accountability; responsibility	Transparency; traceability		Security; resilience; safety		Purposefulness; accuracy; reliability	Lawfulness; respect for nation's values	Explainability
COSO and Deloitte	Accountability; responsibility	Transparency	Fairness and impartialness	Security; Safety	Privacy	Robustness; reliability		Explainability

2.2 Auditability of AI

Auditability of AI is one of the most important dimensions of AI accountability and refers to a set of principle assessments that enable audit procedures to be performed (Liu et al., 2021). Auditability is one of the major drivers of traditional financial and IT audit quality; both financial statement audits and traditional IT audits require auditees to implement auditability measures (Stoel & Havelka, 2021; Stoel, Havelka, & Merhout, 2012). In IT audits, the auditability factor consists of three items: 1) auditees providing competent support to assist in data gathering; 2) well-defined organizational standards and processes (of the auditee) with adequate documentation; and 3) audit team access to unique resources (people, databases, and tools) for specialized audit requirements (Nguyen, Ha, & Nguyen, 2020). Table 2 describes the auditability measures of traditional information technologies.

Table 2: IT Auditability Dimensions

IT Auditability Dimensions	Sources
<i>Robust standards and governance</i>	
Well defined organizational standards and processes (of auditee) with adequate documentation.	Alagić, Turulja, & Bajgorić (2021); Havelka & Merhout (2013); Nguyen et al. (2020); Stoel et al. (2012)
Existence and effectiveness of key controls and their effect on testing.	Havelka & Merhout (2013); Siew, Yeow, Tan, & Grigoriou, 2017
<i>Clients' support</i>	
The audit team is responsive to the auditee's needs.	Alagic et al. (2018); Stoel et al. (2012)
The support given by the auditee.	Alagic et al. (2018); Siew et al. (2017)
Auditee provides competent support to assist in data gathering.	Alagić et al. (2021); Nguyen et al. (2020); Stoel et al. (2012)
Accuracy and reliability of data in system.	Havelka & Merhout (2013)
The employees of client company adequately understood the process and purpose of the audit.	Alagic et al. (2021)
Audit team has access to unique resources (people, databases, tools) for specialized audit requirements.	Nguyen et al. (2020)
<i>Documentation</i>	
Availability and accuracy of documentation for auditability.	Alagić et al. (2018); Havelka & Merhout (2013)
Documentation of the level of automation in business process.	Havelka & Merhout (2013)
Documentation of primary intention and use of system by auditees every day, purpose of the system.	Havelka & Merhout (2013)
System flow charts.	Havelka & Merhout (2013)

Although the current literature on developing auditable AI is largely lacking, auditability has been explored in a few works by researchers and regulators. Shneiderman (2020) distinguished auditability from answerability; whereby the notion of answerability deals with who is accountable, auditability indicates how the designers and implementers of AI systems are to be held accountable. To conduct an audit, there needs to be a set of accessible auditable artifacts that record decisions, systems, and processes during designing and implementing AI (Ayling & Chapman, 2022). Importantly, documentation of model development seems to be a focus of previous work on AI auditability. Multiple studies have recommended the use of model cards and datasheets that make documents auditable (Gebru et al., 2021; Mitchell et al., 2019; Raji et al., 2020). Model cards specify the information on model details, intended use, evaluation and metrics, training and test data, ethical considerations, recommendations, and limitations (Mitchell et al., 2019), and datasheets document the collection process, motivation, composition, and recommended uses of the data (Gebru et al., 2021). Raji et al. (2020) suggested that well-documented model development should include a well-structured and well-annotated codebase, extensive records of the hardware and software used, and explanations of how the model will be maintained once it is put into production. Documentation should also include the purposes and business requirements of the system (Raji et al., 2020). In addition, statistically sound evaluation protocols are also necessary for auditors, through which they measure and assure model quality, and substantiate claims of model performance (Chatila & Havens, 2019). To ensure appropriate documentation of training data, Singapore's PDPC (2020) recommended that organizations maintain a comprehensive record of data provenance, procurement, preprocessing, lineage, storage, and security. This record could also include qualitative input about data representations, data sufficiency, source integrity, data timelines, data relevance, and unforeseen data issues encountered across the workflow. ICO (2020)

extended its scope of documentation to include how risks to the individuals who are having their personal data processed have been considered, the methodology for identifying and assessing trade-offs, the prioritization criteria and rationale for the final decision, and how the final decision fits within the organization's overall risk appetite. Although a few standard documentation practices for AI model development and data process have been proposed in the literature, whether and to what extent these practices have been adopted, and how they contribute to AI system auditability have not been largely verified by audit practitioners.

Ensuring traceability is another important aspect of AI systems (Brundage et al., 2020; Falco et al., 2021; PDPC, 2020; Toader, 2020). Companies should establish mechanisms that facilitate AI systems' auditability, such as ensuring the traceability and logging of AI systems' processes and outcomes (Toader, 2020). Traceability can be promoted by: 1) building an audit trail to document model training and AI-augmented decisions; 2) implementing a black box recorder that captures all input data streams; and 3) ensuring that data relevant to traceability are stored appropriately to avoid degradation or alteration and are retained for durations relevant to the industry (PDPC, 2020). In addition, EU's legislative framework on the free flow of non-personal data also highlighted the necessity for appropriate auditing mechanisms and suggested that competent authorities should be able to take appropriate measures to regulate the process of auditability, such as imposing functional requirements to support access to data and requiring system descriptions (EU Regulation, 2018). However, current proposed measures for AI auditability only target the pre-deployment stage of AI systems. As AI systems can go wrong during and post to the deployment phase, audits need to be continuously and periodically conducted. Thus, the auditability measures needed during and post to the deployment phase need to be studied. Auditability is listed as one of the six dimensions of trustworthy AI ethical principles

(Liu et al., 2021). However, the AI industry lacks proven methods for translating principles into practice, as these principles are written in overly technical language; each principle needs to be interpreted and broken down into specific rules for stakeholders to understand and act on (Raji et al., 2020). AI auditability is important because it holds the key to addressing issues such as AI bias and security (Liang et al., 2021). Auditability is a prerequisite for auditors to perform AI audits; whether and how the AI system is auditable could also affect the workload and quality of AI audit engagements. While the IT auditability dimensions have been well validated in practices and verified by IT audit practitioners (Stoel & Havelka, 2021), studies that systematically address AI auditability measures across the lifecycle of AI systems are lacking. Current proposed auditability measures for AI systems also lack validation by audit practitioners based on their practical experiences.

2.3 Competencies Required for AI Audits

Current literature has not systematically identified the knowledge and skills required for conducting AI audits. Commission de Surveillance du Secteur Financier (CSSF) (2018) suggested that having a team with the right skills can contribute to success in AI projects, given AI project demands for new types of professionals who are skilled in statistics, mathematics, and programming. CSSF (2018) also highlighted the importance of AI knowledge, adequate internal skills to maintain the AI solution, and the involvement of internal auditors who contribute to the success of an AI project. The IIA requires internal AI auditors to: 1) know how AI works; 2) understand the risks and opportunities AI presents; 3) determine whether AI outcomes are as expected; 4) be capable of recommending or taking corrective action if needed; and 5) determine whether third-party providers of AI systems are competent (IIA, 2017). In addition, expertise in audit tools is viewed as one of the most influential drivers of IT audit quality (Siew et al., 2017;

Stoel et al., 2012). AI audits demand a set of tools that are different from, or in addition to, traditional IT audit tools. AI audits can be conducted manually via logical walk-throughs, automatically via code checks, or by large-scale computer simulations (Robert, Pierce, Marquis, Kim, & Alahmad, 2020). Many ethics tools being developed and used mainly target the private sector and are used by the AI delivery, quality assurance, procurement, and development teams; most available ethics tools are used for internal assessment or audit purposes rather than by external reviewers or auditors (Ayling & Chapman, 2022). Liu et al. (2021) also proposed and categorized representative debiasing strategies for AI systems. To assist in compliance with requirements of the act before AI systems can be placed on the market, the “capAI” was created for conducting conformity assessment of AI systems in line with the *AI Act* (Floridi et al., 2022). While both internal and external AI audits can contribute to the trustworthiness of an AI system (PDPC, 2020), external audits are designed to identify AI technology risks externally and serve as accountability measures for these deployed models. The set of knowledge and skills required for external auditors of AI remains undefined.

The complexity and criticality of AI have challenged audit professionals to devote effort and obtain deep expertise from diverse domains. To our knowledge, there has been no study that systematically defines a set of competencies demanded for this multidisciplinary field of external AI auditing. We introduce our research question of the study as follows:

RQ: What auditability measures and auditor competencies contribute to successful audits for AI ethics, lawfulness, and technical robustness?

3. METHODOLOGY

3.1 Data Collection

A semi-structured interview approach was employed to address the research questions. This approach was viewed as the most appropriate way to obtain the in-depth views and experiences of knowledgeable individuals who are intricately involved in AI audits. It was also the most feasible approach when the field of AI auditing was emerging, and AI audit professionals were scarce. Targeted participants include AI audit practitioners, AI specialists, data scientists, data engineers, AI ethicists, researchers, and product managers who have been involved in or exposed to AI audits. The researchers recorded and transcribed online interviews through Zoom or Microsoft Teams to ensure accuracy and completeness in capturing responses. An institutional review board (IRB) exemption was granted for this interview study.

3.2 Interview Questions and Coding of Responses

The interview protocol is presented in Appendix A. Questions 1, 2, and 3 were intended to capture information about the participants' professional backgrounds. In particular, Question 2 provided the participants with an open opportunity to discuss the responsibilities they had in their current jobs; Question 3 was used to capture whether and how participants had been involved in AI audit projects. If participants had served on AI audit teams, they were asked to describe how many years of experience they had in the AI audit profession. Questions 4, 5, 6, and 7 touched on the definition of the AI audit scope and description of AI audit process. In particular, Question 4 was used to capture a general opinion of what AI audits meant to different experts in the field. To elaborate on Question 4, Question 5 intended to have participants describe an AI audit engagement in which they participated or observed. By design, Question 5 ensured that our understanding of participants' answers to Question 4 was not biased and did not cause confusion in the coding phases. Question 6 was designed to capture variances in audit procedures in different contexts; it

allowed participants to discuss what had caused the variances. Question 7 asked about key stakeholders involved in an AI audit engagement other than AI audit team members. This was intended to enhance our understanding of what happened and to whom auditors communicated during an AI audit. We used Questions 8 and 9 to capture information about how AI audit teams should be formed and what expertise areas are demanded for AI audits. Question 10 asked participants to identify the relevant auditability measures that enable AI audits. Participants shared experiences about what was lacking in their previously audited AI systems that made an AI system less auditable; we also encouraged them to talk about what the audited party could have done to enhance auditability. The interview questions were pilot tested with a university professor who specializes in cybersecurity and AI and a knowledgeable AI engineer from a Big Four firm to ensure that they were clear.

Transcripts were independently coded by two individuals: the researcher and an external coder. The external coder was a well-trained researcher who had experience in qualitative studies and majored in both accounting and information science. A coding scheme was developed by the researcher and the external coder to capture the participants' responses based on content analyses of the first three interview transcripts. This scheme was subsequently revised to include additional categories as needed in the researcher's and the external coder's independent codings. Excluding these three responses, the initial overall level of agreement was 81.1 percent. Finally, the researcher and the external coder reached a consensus on any disagreement after careful discussion. The results presented show the final coding.

3.3 Participants

Fourteen of 23 interview participants were directly involved in AI audit projects; on average, they had 6.54 years of experience related to AI auditing and assessments. Among the

participants, 16 were from Big Four accounting firms, while the rest were from other global and local technology consulting firms. Four participants were at the partner or C-suite level, two participants were at the director or head of department level, five participants were at the manager level, four participants were at the senior or experienced level, seven participants were at the intermediate level (with approximately two years or more professional experience but not yet a senior-level expert), and one participant was at the associate level. Traditional financial auditing and IT auditing demand accounting competencies (e.g., accounting standards, risk management, technology and tools, reporting, etc.), business competencies (e.g., clients' regulatory and business environments, industry specialization, etc.), and professional competencies (e.g., ethical conduct, auditing mindset, communication skills, etc.) (American Institute of Certified Public Accountant (AICPA), 2018, 2022). For IT auditing in particular, information technology knowledge (e.g., enterprise resource planning expertise, databased applications, XBRL, information system security, etc.), is considered the most important competency for IT auditors (Stoel & Havelka, 2021). Compared to skill sets of financial and IT auditors, the expertise needed for auditing AI systems consists of AI engineering, data science, subject matter expertise, psychology, and AI risks and ethics (Landers & Behrend, 2023; Raji et al., 2020). Therefore, our sample comprised nine AI engineering experts, two data engineering experts, five data scientists, one AI audit expert with multidisciplinary expertise, one AI architect, one AI ethicist, one AI researcher, two technology risk experts, and one AI product manager. Table 3 provides a summary of the background information for all participants.

Table 3: Summary of Participants

Position	Freq. (%)	Expertise	Freq. (%)	Firm	Freq. (%)	Audit Exp	Freq. (%)
Partner/C-Suite	4 (17%)	AI Engineer	9 (39%)	Big Four	16 (70%)	Yes	14 (61%)
Director/Head	2 (9%)	Data Engineer	2 (9%)	Non-Big Four	7 (30%)	No	9 (39%)
Manager	5 (22%)	Data Scientist	5 (22%)				

Senior	4 (17%)	AI Audit Expert (with multiple areas of expertise)	1 (4%)		
Intermediate	7 (30%)	AI Architect	1 (4%)		
Associate	1 (4%)	AI Ethicist	1 (4%)		
		AI Researcher	1 (4%)		
		Technology Risk	2 (9%)		
		Product Manager	1 (4%)		

Freq.: Frequency

4. RESULTS

We now present the findings of the interviews. We first describe the AI audit scope based on the participants' descriptions; then, we present the findings of the auditability measures that enable audits of training data, AI model, and process and governance. Third, we present our findings on competencies required for AI audits. See Appendix B for a complete listing of salient quotes.

4.1 AI Audit Scope

Participants saw audits of AI systems in different ways. Currently, independent audits of AI systems are performed by external parties or independent departments within the company being audited. Independent audits of AI systems can ensure their compliance with AI principles and ethical standards. The AI principles and ethical standards that auditors check against depend on the existing regulatory standards, internally developed AI principles by the auditing party, or the ethical guidelines that developers follow to develop the AI system. Most participants who served as external auditors of AI had experience in assessing AI compliance against stated technical, ethical, and social standards. One of the other facets that AI audits cover is assessing the correctness and completeness of the results. Though the internal audit function could perform this task to some extent, one of our participants (2320806) believed an external party would do it more efficiently by stating:

Any audit is a process to verify and validate [that] what has been done is of correctness and completeness. An audit for AI is to ensure that the data patterns have been correctly

identified and have been translated into actionable insights for the predictions and analytics. It depends on which industry it is being used and see if internal auditors could cover plenty of it. But, the more efficient identifications of the risks and many other tasks [should] probably be done by external parties.

In addition, independent audits are conducted to assess the processes and governance for the development and use of AI systems. One participant (2320819) stated:

I see [an] audit as a continuous collaboration between an internal department of a company and an external advisory, for instance, that really looks at the processes that are necessary to assess the impact of an AI system, collect relevant information at every phase of the life cycle, so that you're always able to log and consult information, then in retrospect and really understand which decisions were made at which stage that impact the overall performance of the system. So, I see it in a holistic way throughout the whole life cycle.

On the other hand, many AI audits are perceived as internal quality assurance processes, which remains the norm for many firms. For instance, one participant (2320808) stated:

When it comes to audit, we mainly look at the quality assurance perspective. That is, the model working based on our definition of scope and deliverables. Does it solve the problem we initially stated, that is, the purpose of the audit. It's not really an audit; it is quality assurance that is mainly part of our delivery.

4.2 Auditability of AI

4.2.1 Auditing Standards

Current audits and assessments of AI systems are largely based on clients' organizational objectives or industry-specific requirements. Due to underdeveloped universal AI audit standards, many audit firms have developed their own frameworks and standards for AI audits, such as AI

audit templates and some internal toolkits developed for specific AI audit projects at Big Four accounting firms. For example, one participant (2320820) stated:

I haven't found any regulatory guidelines. So, each firm, I think, follows its own guidelines; they develop their own. So our firm has certain guidelines. So I think that's like totally dependent on firms and what they find acceptable actually.

For certain industries, such as healthcare, insurance, and the financial industry, where sensitive data is involved, auditors follow industry-specific standards. The data used for developing AI models for different industries also varies. This results in a variance in audit procedures across industries. For example, one participant (2320804) stated:

It's very industry dependent. I think the general approach would probably be similar, if not identical, within [an] industry. But, if you are going from the pharmaceutical space to financial services, I do think it varies. But, I do think there are a few benchmarks that you assess, regardless of the industry or the nature of the client's work.

Overall, having standards, frameworks, or defined objectives that guide the audit process is the first step to initiating an audit for AI systems.

4.2.2 Documentation

According to 14 participants, audits of the AI systems are enabled by sufficient and appropriate documentation. Documentation was identified as the most important factor affecting auditability across the entire cycle of AI audit, as documentation of relevant processes enhances transparency of the AI system. However, many organizations are not devoting adequate effort to maintaining relevant documentation. Having documentation also saves audit efforts and auditors' time and facilitated auditors' understanding of the underlying system. For example, one participant (2320822) stated:

When it comes to AI, having documentation is incredibly important. If the company has no documentation of how the system was built, it takes a really long time to audit. Because you basically have to go back in history and try to figure it out and piece it together. You have to have the documentation.

4.2.3 Logs

Maintaining appropriate logs is also an essential measure that enables AI audits. Many legacy systems are not auditable because they do not offer functionality, such as logs, to demonstrate the controls. Without appropriate logs, there is no ability to enable tests and the generation of reports. The logs primarily enhance traceability, which is an important aspect of auditability. When it comes to log files, the biggest part is data. Having a “trail-of-data” process and governance was frequently brought up by the participants. When generating logs, some participants believed that the developing party should log as much information as possible about the process, including each step being made in the development, why this step is being made, and who has monitored it. For example, one participant (2320805) mentioned: “Logging each and every update that we are making, and why we are making those updates, and particular disciplinary standards in order to apply the codes as well, is also important.”

However, some other participants advocated for a balanced approach to preparing logs. One participant (2320811) stated:

The logging depends on what your model is supposed to do; it is not necessary that every decision to be logged. The log data could be too much, and not each data is super useful because you have to do a whole set of post-processing afterwards. But, on the other hand, if you have no logging information, the audit also becomes impossible because you don’t get any metrics outside of the outputs. You need to have appropriate loggings; that means

if you have a high-quantity model, you make sure everything that needs to be monitored is logged. You will choose a logging model that is appropriate for the model and for future monitoring and auditing; it catches the performance metrics that are necessary for anything that is strange and adverse. The audit efforts are put into anomalies that have been detected, which is a reasonable amount of data to look into but also results in statistical significance.

4.2.4 Clients' support and competency

Regarding clients' personnel where AI is applied, key management involved in AI governance must understand where and how AI is being used across the enterprise. In particular, the development team and managers involved should be knowledgeable about the underlying model and the data used. Developers and managers of the project should understand the compliance requirements well, such as data privacy standards, data security concerns, AI regulations, and standards, to make sure the developed AI adheres to the compliance requirements. The developers should be sufficiently technical and competent in coding languages to reduce the technical debt (i.e., potential cost of future reworking needed when choosing a limited and simple solution for the code instead of a better but more time-consuming approach) that has caused extended auditors' time and efforts.

4.2.5 Model intended use

Understanding the intended use of an AI system is an initial step in AI audits. While having well-documented intended use is critically important, organizations usually fail to do so. They also tend to miss proper documentation of the AI system scope and the domain companies to which the AI system applies, which has caused wrongful uses of AI. Well-documented intended use is especially important for AI systems that deal with highly sensitive data. Good documentation of the intended uses should enable auditors to understand the whole context in which AI is used. By

understanding the intended use of the AI system, auditors can begin to understand why and how it causes biases. Documentation of intended uses in the consulting world is often in PowerPoint. To better understand the intended use cases, our participants have suggested many alternative approaches, such as holding knowledge transfer meetings for auditors, or having auditors access meeting minutes between stakeholders. For example, one participant (2320808) stated:

If you understand the intention of the design, whether with very detailed documentation, or the knowledge transfer meetings, if you have [a] sort of detailed pre-context on what it is trying to do, and the reason for the project, this kind of background context really helps you to evaluate the algorithm.

4.2.6 Model explainability and transparency

Explainability and transparency have been huge challenges in the AI profession; they have also caused difficulty for AI audits. According to participants, non-black box model is ideal; however, there is always a trade-off between complexity and explainability. Many participants rated the lack of explainability as the biggest barrier to achieving auditability because complex models like neuro networks often fail to explain themselves. Neuro networks lack explainability, and there have not been many ways to effectively overcome it. Our participants suggested two ways regarding the implementation of explainability in AI systems, particularly for non-black box ones. The first approach is to have comment sections on the AI algorithms. A participant (2320805) stated:

Comment sections about each step that AI is doing as much as possible so that we can interpret. It is not necessary that we'll be able to interpret each and everything out of it. Sometimes it is like a black box, but at other times it's just plain and simple. We can reduce our model in such a way that it is auditable, and it's simple to understand by anyone.

The other approach is to use tools that automate and track information on AI model performance and individual users' behavior within the system, such as AI dashboards. One participant (2320823) stated: "... the auditable AI dashboard should be designed to ensure model explainability or the overall explainable AI."

4.2.7 Model selection and model values

Decision-making on model selection and the value of the model needs to be documented, or informed through auditors' meetings with data managers, product managers, and developers. Model values, including technical advantages, qualifiable and strategic benefits, monetary benefits, should be provided to the audit team. During model selection, the process of comparing candidate models and the selection criteria should be documented. Auditors should be informed of the decision-making process of model selection and participated personnel. In addition, to inform what the model is, the audited party needs to describe the level of automation for the underlying model. The audited party must justify the choice of the model and describe the process of making the choice. Auditors also need to know the value of the model, as stated by an participant (2320802): "...what are the contributions...what [are] readily available market products which can replace their existing models; whether the model could be beneficial and whether they can have a third-party model rather than building their own custom models."

4.2.8 Model operations

Model operation evaluation is usually involved in the quality assurance process. This is enabled by providing descriptions of the model operations. To evaluate the model operations, auditors need to investigate how the model performs; this extends from model development to the use of the model, as one of the participants (2320815) stated: "What is their model? How was the

model performing? And not just when it was built, but also on an ongoing basis, to make sure it doesn't deteriorate in quality over time."

One of the key elements of model operations is causal relationships, and participants suggested documentation on about these relationships should be as detailed as possible. For example, one participant (2320813) stated:

... You need to have detailed maps to show all the causal relationships that happen even before you get to build the AI system ... being as detailed as possible ... we are basically looking at all the different flows, and how all the causal relationships go together.

What also matters is the historical operation of the model based on tests being done throughout the development phase. Recorded historical operation results are usually obtained directly from meetings with the development team and the management persons who were involved. For a data scientist to know whether AI is behaving as expected, they must be able to access operation metrics and make recommendations to take corrective actions. Accompanying the historical operation results, the defined operation metrics and management's steps for addressing any issues from the results also need to be provided. For example, one participant (2320817) stated:

Does the organization have a culture of risk management or not, because if it does, we expect to see things like risk forums; and depending on which, someone will have done a risk assessment on the model and the product typically that it's embedded within, and have a set of again defined thresholds for what it looks like from a performance point of view, and there would be regular testing of the model within that. So, I would be looking for the presence of those kinds of artifacts. Is there testing being done? What are the results saying? What is management doing based on the results that are coming back? How are they

identifying and addressing any issues that are being identified. And if those things aren't in place and documented, then it's very difficult to justify that there's a good governance structure in place.

4.2.9 Model deployment and monitoring.

During model deployment, the underlying AI model is put into production. Model deployment is the process of putting AI models into production, where the model's outcomes become available to users. Post to model deployment, the AI model continue to be monitored (e.g., quality assurance, updates, etc.) to maintain high model performance. It is important not only to document how the model is deployed but also how it is monitored over the lifetime of the AI system. For example, one participant (2320811) stated:

Another important aspect is providing details on how the model is embedded into the overall product and process. It is not only the model performance itself, but also how the model is monitored, how you measure the performance of the lifetime of the product, and what kind of measures have you foreseen in terms of retraining the model and figuring out when your model fails.

4.2.10 Training Data accessibility

Auditing training data is an essential part of the AI audit process. Hence, the audit team needs to access training data. Availability and accessibility were identified as key elements of data auditability. For example, one participant (2320812) stated:

The most important thing with AI in general is that they forget about the importance of clean data; you have to be able to know the data it's working with. The hardest part for us to implement an AI audit is being able to have the data.

According to the interview results, auditors did not need all training data used. Some samples of the data must be provided for auditors to determine if there is something adverse in the training data. In addition, it must be easy enough to pull out the needed data to save auditors' time and effort.

4.2.11 Data characteristics

While not all training data are needed for an audit, the description of the data was identified as an important factor that enables the audit. Description of data characteristics should cover information essential for understanding the data, such as data creation, data source, data subjects, variables, etc. To help auditors and other potential stakeholders understand the datasets used to train the AI model, the features and uses of the training data should be well captured. For instance, an participant (2320811) stated: "We all know the training data itself and the AI model can be extremely extensive; therefore, it has to be described in a way that all the data features and uses have been captured."

In addition, quality measures for training data should be well developed and stated, especially when the data are purchased from external providers. For instance, one of our participants (2320819) stated:

What is essential is to even find quality measures, to develop methodologies that can be implemented to actually assess the quality of the data you're feeding to the model, because currently there is a lot of discrepancy. So, for example, some companies buy data from an external provider, but they don't have any methodology to actually assess the quality of this data that they're feeding to the model to be developed.

4.2.12 Model governance

The governance information surrounding the model is also required for an audit. This governance information should also inform the procedures to be taken if the model fails. For example, one participant (2320817) stated:

We would also want to be looking at not just the model itself but all the governance structures around it. So, how is it being fit? What happens with the information that comes out on the other side of it? How is it being looked after, governed, managed, and restricted appropriately.

In addition to the stated governance standards, the audited party should also provide any updates to the governance standards and continuous assurance policies. For example, one participant (2320818) stated:

Is there a good governance standard? So, this kind of governance, and is that being updated?

I think, on top of that, what is your continuous assurance and continuous compliance model? ... If it's documented well enough, auditing gets much easier because you have something to audit against.

4.2.13 Data governance

Within the overall process of training AI systems, documentation and logs that reflect process around data should be maintained. Auditors should be informed of how the data was sourced, evaluated, cleaned, used, and stored. Data privacy, security, and the quality of training data need to be well-governed. There should be robust policies present on privacy and security aspects, either the external standards that the developing party followed, or the internal rules and criteria used to govern data. For example, one participant (2320806) stated: "The data governance should be very robust, which means the data quality, data security, data trail, and data compliance

should be adhering to the whole set of policies and security aspects. This will make the audit process much more systematic.”

4.2.14 Governance structure

Governance structure needs to be well documented. Such governance involves process governance around model development, data governance, as discussed earlier, and the management governance structure. However, the governance structure information is often not well described. The management governance structure provides auditors with information about key stakeholders’ roles in the AI designing, developing, deploying, and operating process and their respective responsibilities. Documentation of governance structure should also include people who are responsible for each phase, people who authorized different deliverables, and the decision escalation process from developers to C-suites. For instance, one participant (2320810) stated:

Who is authorizing all those deliverables, and how does that authorization get escalated from data scientist to leader and then CEO, CXO? What is the escalation process, and what are the controls? So, all of that is what makes a program successful, not just having a few people code.

4.2.15 Summary for AI Auditability

In summary, there are five major components of AI systems, including training data, model, process, governance, and people. Auditable training data requires data accessibility and information about data characteristics and quality assurance; auditable model requires documented model intended uses, model selection and values, model operations, and model deployment and monitoring, and enhanced model explainability and transparency; auditable process requires sufficient logs throughout the AI lifecycle; auditable governance requires clear external and internal rules that govern the process and quality of AI; and auditable people involves a governance

structure that details key stakeholders' roles in developing, deploying, and monitoring the AI system. Each of the five AI components has demanded a variety of auditability characteristics to be implemented. Based on the interview results, we specify required steps to enhance auditability for each AI component to be audited. Table 4 displays our proposed auditability framework for AI systems based on our findings. Table B1 lists the key themes of AI auditability and all supportive qualitative evidence.

Table 4. A Summary of AI Auditability Results

Component for Audits		Auditability Measures
<i>Training data sets</i>		
1.	Data accessibility	• Ensure availability and accessibility of training data.
2.	Data characteristics	• The description of the data, including the features and uses of the training data, should be well captured.
3.	Data quality	• Ensure the quality of training data, and provide quality metrics used to measure data quality.
4.	Security and privacy considerations	• Security and privacy considerations need to be given early during model design and development.
<i>The underlying model</i>		
5.	Model intended use	• Well-documented intended use including the AI system scope and the domain companies and contexts to which the AI system applies.
6.	Model explainability and transparency	• Implement explainability features in the model. • As the black-box nature is not acceptable, well commented sections on AI algorithms, automated information tracking on model performance and user behavior within the system, or other methods that enhance explainability can be implemented to explain highly complicated models.
7.	Model selection and model values	• Decision-making on model selection and the value of the model needs to be documented. • The audited party needs to describe the level of automation for the underlying model.
8.	Model operations	• Description of the model characteristics, model performance, and the performance evaluation metrics on an ongoing basis. • Description of the causal relationships. • Records of historical performance results and risk analyses results of the model throughout the development phases.
9.	Model deployment and monitoring	• Documentation of how the model is embedded into the overall product and process. • Documentation of how the model is monitored over the lifetime of the AI system. • Description of the measures expected to be used in terms of retraining the model and figuring out when the model fails.
<i>Process, governance, and people</i>		
10.	Data process and governance	• Documentation and logs should be maintained for how the data was sourced, evaluated, cleaned, used, and stored. • Data privacy and security need to be well-governed with robust external privacy and security standards or internal rules and criteria. • Quality measures for training data should be well developed and stated, especially when the data are purchased from external providers.
11.	Management governance structure	• Well-documented governance standards and continuous assurance and compliance model. • Documented information about management governance structure. Documentation should include people who are responsible for each phase, people who authorized different deliverables, the decision escalation process from developers to C-suites, and the performance metrics and thresholds.

12. Model governance	Well-documented governance structures around model development, including how the model is being selected, fit, managed, restricted, and deployed.
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4.3 Competencies Required for the AI Audit Team

4.3.1 Technical Competency

The audit team needs to possess a combined set of knowledge and skills to conduct successful audits of AI systems. The key technical knowledge areas demanded for AI audits are AI engineering, data engineering, data science, software engineering, and knowledge about the underlying model being audited. Coding competencies are required along with each demanded expertise area. Such technical competency is needed for understanding the underlying model and data, designing and running audit tests, and interpreting and evaluating test results during the overall audit process. The required expertise in each of the above-mentioned areas is at a very high level. For instance, one participant (2320811) stated:

When it comes to assessing the AI model, you really need quite deep expertise on how AI works in general ... The profile would be data scientists that are very technical, computer scientists that have quite deep knowledge of AI models.

Sometimes, auditors are even required to be better experts in certain types of models to be able to audit the model. For example, one participant (2320807) stated: “Auditors might even have to be better deep learning experts than the people creating those models and applications themselves.”

While it is not necessary for each team member to be an expert in AI or data science, basic knowledge of AI is required for everyone on the auditing team. For example, one participant (2320801) stated: “It’s important for all team members to know basic theoretical knowledge, be able to explain what artificial intelligence does.”

4.3.2 Compliance Knowledge

Knowledge about compliance requirements and legal aspects is an influential factor in AI audit success. Such knowledge is heavily demanded for determining the audit scope and assessing AI risks. Auditors need to know about the regulations and standards that regulate AI and guide the AI audit process. As regulations are emerging in this field, being knowledgeable about AI regulations has been identified as an increasingly important factor. For example, one participant (2320805) stated: “We are to make sure that none of the regulatory standards are compromised, so we need to have a knowledge of that side as well.”

Having a team member who specializes in compliance requirements (e.g., regulations, auditing frameworks, etc.), sometimes referred to as the auditor or a compliance subject matter expert, becomes a necessity. Regulations and standards surrounding data were perceived as particularly important. For example, a participant (2320805) stated: “We are to make sure that none of the regulatory standards are compromised, so we need to have a knowledge of that side as well.”

As of now, many audits of AI systems have been subject to industry-specific standards, leading to a demand for subject matter expertise. Being an expert in industry standards was discussed as one of the key challenges for AI auditors. When the audited party has developed and followed its own standards for AI, auditors should also understand their internal rules and compliance functions. Another key skill for AI auditors is to translate these compliance requirements into actionable measures. For example, one participant (2320819) stated:

You need to translate some ethical requirements or legal requirements in the case of the AI Act that are coming into mathematical formulations into code ... and how you can effectively communicate with developers into translating such requirements into practices into actionable measures.

4.3.3 Business and Process Knowledge

Understanding the business process of the company to which the AI system is applied is another key expertise required for AI audits. This also emphasizes the importance of having a subject matter expert on the audit team. Auditors should not be limited to model development but also deployment and uses. Auditing the business contexts requires auditors to know the users and how they used the model. Knowing the business contexts to which the application is applied and the business users is critically important for analyzing AI risks and impacts. For example, one participant (2320810) stated:

One is that they need to understand the business as to what the business purposes, how the models are being used, what the users are, and how they are using the model. I think that's critical. So, business knowledge plus data science knowledge would be the key.

4.3.4 Auditing Skills

Some skills and mindsets traditionally required for the audit profession have also shown some impact in the AI audit field. Such traditional auditing skills (e.g., computer assisted audit techniques, etc.) remain important during the audit process of evaluating audit evidence and writing audit reports. Having a mindset for auditing remains important in AI audits. One participant (2320815) stated: "And the fifth is the mindset of validating and auditing."

4.3.5 Competency in Audit Tools

Competency in using audit tools for AI systems was also identified as an important skill by the participants. Auditors must know and be able to use advanced audit tools to collect audit evidence in AI audits, measure and assess AI systems.

4.3.6 Knowledge of Social Implications

The auditors on the team need to understand the social implications of AI risks. This knowledge becomes particularly important when assessing AI risks and impacts. Sometimes, the audit team needs to hire a social scientist to identify biases and risks. For example, a participant (2320807) stated: “And finally, you would definitely need a sort of social scientist person depending on the obligation. That person is responsible for identifying biases against specific groups, minorities, and other things.”

The individual tasked with examining the social implications of AI systems, sometimes called outcome experts, can be a high-level executive who has a big picture of the product and its users. For example, one participant (2320814) stated:

The third one, I would say, it’s someone who understands the outcomes of this, and can understand what would actually happen if we predicted a certain way or not. If there’s bias within our model, and we’re not auditing our models correctly, what are the risks. The outcome expert, or the data engineer somewhere on the user side. To me, this is someone more like a higher-level executive who has knowledge of a product that you’re building. They understand their users. They understand the use case that this is meant to do.

The examination of social implications of AI systems also calls for the involvement of psychologists. A participant (2320819) stated: “There should be psychologists. I see a role emerging because we are interacting more and more, we are exposing children to these technologies, so really preventing harmful content, these are big challenges that need to be addressed.”

4.3.7 Soft Skills

At the end of the audit, auditors must be transparent with their clients and public audience and be able to interpret audit results in plain language so that the intended audience can understand

them. This calls for storytelling skills and concerns for social good. For example, one participant (2320823) highlighted storytelling skills for technical team members:

If the AI engineers are working with the product owner or case masters or with the stakeholders, then a skill will be required in storytelling. Being able to take those complex statistical outcomes or mathematical outcomes and translate them into a business language that can be understood, that is where a lot of the failures also happen.

In the meantime, auditors on the team need to have social goodness and truthfulness. An participant (2320809) considered social goodness the first requirement for AI auditors:

The passion, the motivation, the whole thing should be put into any AI auditing. The whole thing is social goodness. The first and primary concern of anyone who does it is whether that person is socially good or not. That is the first thing. The second thing is faithfulness: Are we faithful to people who are concerned. This is more than any technical expertise.

4.3.8 Summary for AI Audit Competency Requirements

In summary, as audits of AI systems are co-efforts across broad domains, a comprehensive audit team should exhibit all the core competences demanded. Table 5 displays the required knowledge and skills for conducting AI audits and the number of participants who mentioned a particular required knowledge or skill. Table B2 lists key themes and supportive qualitative evidence of the required competencies for AI audits.

Table 5. Knowledge and Skills Required for AI Audits

No.	Knowledge and Skills Captured	Freq.	Description
1	Knowledge of compliance requirements	13	AI auditors should be knowledgeable about the compliance requirements for AI
2	AI knowledge	11	AI auditors should have AI knowledge
3	Knowledge of the underlying AI model being audited	9	AI auditors should be knowledgeable about the underlying model being audited
4	Business knowledge	9	AI auditors should be knowledgeable about the client's business process
5	Data science expertise	8	AI auditors should have expertise in data science, statistics, and mathematics
6	Data engineering expertise	6	AI auditors should have data engineering expertise, including database and systems, data modeling, data sampling, and data analytics

7	Knowledge of social implications	5	AI auditors should have a good understanding of social implications of AI technologies
8	Auditing skills	5	AI auditors should understand the auditing process and have auditing skills for AI audit projects
9	Coding competency	4	AI auditors should have strong coding competency
10	Competency in AI audit tools	2	AI auditors should know how to use audit tools in AI audits
11	Software engineering skills	2	AI auditors should have software engineering skills or experience in developing software
12	Soft skills	2	AI auditors should have social goodness and storytelling skills

Freq.: Frequency – the number of participants who mentioned the corresponding knowledge and skills

5. DISCUSSION

According to the current state of AI audit practices reported by our interview participants, AI audits are categorized into four categories: 1) ethics-based audits of AI bias, fairness, interpretability, explainability, privacy, security, and safety; 2) compliance audits against policy and regulations during which auditors evaluate whether AI systems are in compliance with relevant regulations or industry standards; 3) process and governance audits throughout AI design, development, deployment, and operation phases; and 4) quality assurance that evaluates results reliability, correctness, and accuracy. In contrast to earlier trends where AI audits were mostly conducted internally (Ayling & Chapman, 2022), now Big Four accounting firms and third-party technology consulting firms are increasingly involved in performing external AI audits and assessments. It is generally acceptable for audits focusing on technical robustness to be done internally, however, compliance audits and ethics audits are expected to be done by external third parties. This aligns well with Shneiderman's (2020) assertion that the assurance of AI trustworthiness serves as the third layer of governance conducted by external review organizations in addition to AI reliability and safety layers, which are assured by internal reviews. There are well-defined practices for internal financial and IT audits, as well as how external auditors can use internal audit reports during their audits. Future research is necessary to articulate internal audit practices for AI and how these can interface with external AI audits.

Current AI audit practices lack universal guiding standards and scope requirements; third-party auditing firms, such as PwC (PwC, 2019) and Deloitte (Deloitte, 2020), have developed and followed their own frameworks to help clients design and deploy trustworthy AI. For development of regulations in the AI space, an articulation of risks of AI systems and their mitigation strategies must be clearly identified. In addition, current audit practices for AI systems generally focus on bias and discrimination, fundamental human rights, invasions of privacy and unreliable, unsafe, or poor-quality outcomes; however, other potential risks of AI systems, including the lack of transparency and explainability in AI and the loss of social trust, empathy, and mutual understanding due to excessive automation (Leslie et al., 2019) have not been sufficiently addressed in the current AI audit practices and need to be considered in the future. Current guidance on AI audits is voluntary (e.g., NIST's AI RMF); many audits, or assessments of AI are currently done as voluntary self-regulation without external governance mechanisms. This calls for future regulatory efforts of AI systems to develop universal standards or benchmarks for compliance and ethics audits for AI systems (Schöppel et al., 2022).

We propose an auditability framework for AI that guides AI developers and operators in implementing auditability measures for each component of AI system to be audited, including model, data, and process and governance (Figure 1). At the model level, auditors need to know intended model use, model values and selection criteria; model operations, and model deployment and monitoring; the model needs to have explainability and transparency measures, especially for complex and hard-to-explain models. At the data level, the audited party must ensure auditors' accessibility to training data and provide any necessary information on data characteristics and provenance; the quality of the training data must be ensured, and the quality metrics needs to be provided to auditors; security and privacy considerations must be given during the design and

development stages. At the process and governance level, documentation and logs that indicate governance standards and process for training data and model building must be provided to auditors; auditors also need the governance structure information to determine who is accountable.

In addition to auditability measures at the component level, we provide general practices that enable AI audits as a whole. First, a robust standard and governance must be in place for auditors to audit against. This aligns with many responsible AI studies that call for robust governance for AI development and AI ethics (Stahl, 2022). This standard can be a universally accepted regulation or a standard applicable to a specific industry. When the regulatory and industry standard are not in place, the auditing party may develop their own toolkits derived from robust research and field experiences. Second, clients' (i.e., the audited party) key stakeholders must be knowledgeable about process and governance related to the development, deployment, and operations of the AI system; they also need to actively support the audit team by providing all demanded materials during the audit. Third, documentation serves as a tool to support auditability measures at the model, data, and process and governance levels. Fourth, appropriate logs need to be maintained that capture details of inputs, outputs, data processing and computation, and any actions taken to give effect to the decision; in addition, process logs that record process events, such as changes in state, warnings, or errors, should also be provided. This framework is intended to be used by AI designers, developers, and operators to implement appropriate measures that support auditability.

Figure 1. AI Auditability Framework

Robust Standards and Governance		Clients' Support and Competency							
- Universally accepted AI and data governance or industry-specific regulations - Internally developed toolkits by audit firms		- Accessibility to demanded materials - Stakeholders' knowledge about process and governance							
<table><tr><th>Model Auditability</th></tr><tr><td> - Model intended use - Selection criteria and values - Model explainability - Model transparency - Model operations and causality - Model deployment - Long-term monitoring </td></tr></table>	Model Auditability	- Model intended use - Selection criteria and values - Model explainability - Model transparency - Model operations and causality - Model deployment - Long-term monitoring	<table><tr><th>Process and Governance Auditability</th></tr><tr><td> - Data process and governance - Model governance - Management governance structure - Any updates on governance </td></tr></table>	Process and Governance Auditability	- Data process and governance - Model governance - Management governance structure - Any updates on governance	<table><tr><th>Data Auditability</th></tr><tr><td> - Accessibility to training data (or samples of training data) - Description of data characteristics and data provenance - Data quality measures - Security and privacy considerations </td></tr></table>		Data Auditability	- Accessibility to training data (or samples of training data) - Description of data characteristics and data provenance - Data quality measures - Security and privacy considerations
Model Auditability									
- Model intended use - Selection criteria and values - Model explainability - Model transparency - Model operations and causality - Model deployment - Long-term monitoring									
Process and Governance Auditability									
- Data process and governance - Model governance - Management governance structure - Any updates on governance									
Data Auditability									
- Accessibility to training data (or samples of training data) - Description of data characteristics and data provenance - Data quality measures - Security and privacy considerations									
Available and Accurate Documentation		Sufficient, Reasonable, Appropriately Stored Logs							
- Model - Training datasets - Process and governance		- Logs for model training - Logs for input data streams and AI-augmented decisions - Logs for data process and governance							

Based on audit practitioners' domain experiences, this proposed AI auditability framework verified many key auditability measures suggested in the previous research, including documentation and log practices for model development, model evaluation, and data processes. The framework also incorporated traditional IT auditability measures that are found useful in the context of AI audits, such as having well-stated compliance standards, internal governance, and clients providing access and support for auditors in information gathering (Alagić et al., 2021; Havelka & Merhout, 2013; Nguyen et al., 2020; Siew et al., 2017; Stoel et al., 2012; Yeghaneh, 2015).

Compared to the traditional auditability requirements for IT audits, well-documented intended use and model deployment have become critically important in the context of AI, as AI applied to unsuitable business contexts could cause severe social implications. Compared to IT system documentation requirements, documentation of AI models involves documenting model values and model operations. One key element that distinguishes AT systems from traditional IT systems is that AI is trained with datasets, which potentially increases AI's risk of inheriting

societal biases. Therefore, AI audits demand documentation of training data, which includes the documented data features, data processes, quality assurance, and governance. Process and governance surrounding model development and training datasets are highly demanded in governance and process audits for AI systems. While regulations and governance for IT audits are relatively mature, governance for AI audits may be lacking in many countries and industries and may vary across organizations, industries, and countries. Therefore, AI developers and implementers must explicitly define the governance rules they used over the lifecycle of an AI system. In addition, making timely updates to the governance standards and providing the update information to auditors is also important for AI audits. Such documentation could be compliance document that breaks down each high-level principle into more tangible and actionable formulations (Mökander & Floridi, 2022). Auditors also need to be provided with information about governance structure, so key stakeholders can be held accountable. The documented governance structure has not been emphasized for IT auditability.

As a unique contribution, this study also identified additional auditability issues and measures that have not received attention in the existing literature. First, documentation should be maintained for the value and selection of the underlying AI model; justifications of the model choice and alternative choices available should be well informed. While our study concurs with the documentation and log practices on model performance and performance metrics proposed in the existing literature, our findings reveal that a detailed map of the causal relationship will greatly enhance auditors' ability to understand and audit each decision or prediction made by the system. Auditability can be enhanced by understanding why a model made a specific prediction (Maughan, Ngong, & Near, 2022), understanding causal relationships between variables also help enhance the explainability and interpretability of the model. This becomes extremely important for auditing

black-box models. However, the development of detailed and understandable causal relationships is often missing in the current auditability practices. Third, AI deployers and operators must document processes and governance surrounding AI deployment and monitoring, such as how the model is fitted into the overall product and process and long-term quality assurance process. Fourth, the governance structure, such as a description of roles involved in designing, developing, deploying, and monitoring an AI system, is informative in determining who is accountable. Fifth, in addition to documenting data features and processes, the accessibility and the ease to access the training data poses challenges for AI auditors. What facilitates an effective and efficient audit is the clean and accessible training data, at least sufficient samples of training data. This also leads to a need for future regulations to accommodate auditors' access to necessary training data, especially those involving sensitive personal data and subject to personal data protection. Sixth, our findings reveal the key obstacle of lacking quality measures for input data, especially for data purchased from external providers. Future work should aim to define valid and practical measures that serve as essential pieces of data governance. Seventh, echoing recent AI audit research (Falco et al., 2021; Shneiderman, 2020), maintaining appropriate logs is also an essential step toward AI auditability. While some practitioners demand logs to be created for each update that is made, auditors only need a reasonable number of logs to examine but also result in statistical significance. Future work should aim to suggest appropriate logging models that capture necessary information across the lifetime of an AI system to accommodate AI audits. Lastly, given the black box nature that many AI models are subject to, explainability has been identified as an important antecedent of auditability. Explainability ensures that outcomes are explainable and interpretable to stakeholders, therefore viewed as a tool for evaluating the soundness of the model (Bhatt et al., 2020; Toreini et al., 2020). Our interview participants recognize many tools that could be adopted

to track AI behavior and usage over time, such as ModelOps tools and AI dashboards. Therefore, organizations should adopt these tools to provide the ability to examine the models in depth. Future research may explore the differences in auditability measures that are suitable for explainable and black-box AI systems. Developing advanced AI tools that could assist in AI system audits, particularly for black-box and other complex AI systems, can be a promising future research direction.

Having the right expertise on the audit teams is a prerequisite for a feasible and successful AI audit. Our study identifies a broad set of knowledge and skills across multiple domains that are necessary for AI audits, including, IT auditors, AI engineers and architects, data scientists, data engineers, compliance experts, domain or subject matter experts, business and industry experts, social scientists, psychologists, and lawyers. Among the demanded expertise, knowledge of compliance requirements and knowledge of AI systems are the most frequently discussed items by our participants. Stahl (2022) argued that the developers of AI systems should have conceptual knowledge of reasonably foreseeable ethical implications; our results show that auditors also need to understand social implications and obtain socioeconomic experiences. The need for psychologists in the audit process concurs with Landers and Behrend's (2023) call for psychological perspective on measurement and human characteristics to be integrated in AI audits. In line with previous research (Floridi et al., 2022; Leslie et al., 2019), the ability to translate high-level ethics principles into actionable measures or verifiable criteria is essential for both AI development teams and AI audit teams. In addition, traditional auditing skills, such as the auditing mindset, communication skills with stakeholders, and ability of using audit tools, continue to be demanded in AI audits. Current audit tools are largely used internally as self-assessments, with a focus on appropriate documentation for verification and assurance, as well as having appropriate

governance mechanisms in place (Ayling & Chapman, 2022). To fulfill the objectives of external AI audits, external AI auditing providers should be trained in AI audit tools, such as debiasing tools (Liu et al., 2021; Zhang et al., 2018), ethics toolkits (Wong, Madaio, & Merrill, 2023), and emerging compliance assessment tools (Floridi et al., 2022). Future research should continue to advance the usability and quality of AI audit tools. Auditors also need to continue to be trained in emerging and advanced audit tools to improve audit effectiveness.

Overall, based on the auditability measures identified in this study, future research should quantitatively examine the relative importance of proposed auditability measures and how they impact AI audit success. As auditors' personal and interpersonal competencies have traditionally affected audit quality in the financial and IT audit quality literature (Knechel, Krishnan, Pevzner, Shefchik, & Velury, 2013; Merhout & Havelka, 2008), future research should examine how required competency factors for AI audits affect AI audit quality.

AI models train based on existing data and can map the input to the output over complex hyperplane. Over time AI learning models are getting stronger in conjunction with improvement in our computing capacity. Large language models are the most recent innovation in the field of AI and their evolution has serious implications for the auditing world. GPT had 117 million parameters, GPT-2 had up to 1.5 billion parameters and GPT-3 had up to 175 billion parameters. Unlike conventional information systems where software can be tested for correctness of the algorithms. AI based systems can only be tested stochastically through a sampling of input and output. As models become complex; it will be hard to identify the regions of the hyperplane where the model is underdeveloped. Topic focused language models are being developed in specialized fields such as healthcare, and we expect one will be developed for accounting. As accountants use AI for their work and auditors rely on these models to support audits there is a chance that

systematic errors will creep up that will go undetected. Moreover, as AI systems evolve we may give them the ability to adapt for improved efficiency – how do we audit a system that is morphing on its own? We need to comprehensively examine failure modes for these systems and develop detailed fault trees for AI systems and create standards for developing and testing them. There is a great need for future regulatory and research efforts to address the implications of AI misuse and measures to mitigate harms. Another area where organizations need to be more vigilant is AI loafing where employees leverage AI for their work and engage in their personal pursuits, instead of focusing on official work. We believe that organizations should embrace generative AI for performing routine and repetitive tasks to both improve efficiency and quality of work for employees, they need to have policies that clearly delineate where it is appropriate to use generative AI and where it is not.

5.1 Implications

Our study serves as the first formal attempt to systematically identify and classify auditability measures and auditors' expertise demanded for AI audits. Our findings contribute to the literature uniquely by providing perspectives from audit practitioners, as the auditability of AI systems directly impacts and facilitates the work of audit practitioners. Using qualitative interviews with field experts, our findings enrich the literature body by defining auditability measures for AI systems and distinguishing it from the traditional auditability measures for IT audits. We also identify a set of knowledge and skills required for AI audits across domains. As system auditability and auditors' personal and interpersonal competencies have been identified as quality drivers in the IT audit quality literature (Merhout & Havelka, 2008; Stoel & Havelka, 2021), understanding AI system auditability and AI auditors' required personal and interpersonal competencies is a crucial step in enhancing AI audit quality. Having detailed mapping of

auditability measures and auditors' competency dimensions will also facilitate AI audit quality research and provide resources for measuring AI system auditability and auditors' and audit team's competency levels.

The findings of the study also have implications for the accounting information system domain. Various AI systems have been applied to advance accounting information systems (e.g., Qasaimeh et al., 2022; Zhang, Cho, & Vasarhelyi, 2022); such AI-powered accounting information systems need to be scrutinized both by the business and the auditors. The findings on AI auditability measures and auditor expertise for AI audits based on AI systems applied in broader domains can be applied to the audits for accounting information systems, to some extent. According to the risk categories proposed by AI Act, AI solutions embedded in accounting information systems are likely to be considered as having limited risk or minimal risk to human safety (EU, 2022). Current audits on limited risk or minimal AI systems are largely voluntary (e.g., EU, 2022; NIST 2022), sometimes subject to the desire of the developing party or the request by the acquiring party. Such voluntary audits concern more about AI quality, which in the context of accounting information systems may mean the technical robustness and results accuracy, data privacy, security, accessibility, and misuse, model explainability, transparency, and accountability. These potential audit objectives for AI-powered accounting information systems are in line with the common ethical risks in the development and use of AI in managerial accounting (Zhang et al., 2023). The auditability measures for training data may be particularly relevant for auditing AI-powered accounting information systems, as the data quality-related risks in accounting information systems can significantly impact the business decision-making and regulatory compliance (Bai, Nunez, & Kalagnanam, 2011). Future research may design auditability measures for auditing the quality and quality assurance criteria of the data used to train AI-powered

accounting information systems. To facilitate audits of financial systems, auditability measures used for AI governance and process and AI technical robustness may be more relevant in the context of AI-powered accounting information systems. For financial system auditors, the theoretical knowledge of AI, data science, and data engineering, knowledge of the underlying model being audited, and competency in AI audit tools may be demanded for auditing AI-powered accounting information systems, in addition to traditionally required accounting and information system expertise (Askary et al., 2018). As a limitation of this study, we did not specifically investigate the auditability measures and auditor expertise for auditing AI-powered accounting information systems. This may be a promising future research direction.

Our research also has many practical implications. First, we provide an overview of what is being done in the AI audit profession to address AI lawfulness, ethics, and technical robustness. While current research work is emerging, few have obtained perspectives and insights from current AI audit practitioners. Knowing the current state of audit practices has led us to identify gaps in the field, such as the lack of a sufficient workforce and appropriate expertise in the AI audit profession, the lack of universally accepted regulations and compliance requirements, explainability issues, the lack of quality measures for training data, unclear categorization of AI systems based on risk levels, and gaps in translating high-level principles into actionable measures. The identified gaps need to be addressed in future research.

Second, we propose a framework for AI auditability that provides a detailed mapping of the auditability measures to be implemented for each significant component and process of AI. Future research may use our AI auditability framework to spur AI design that support auditability. The adoption of auditability measures and the level of implementation should be determined based on the potential risks of implementing the AI system, complexity of the underlying AI project, and

the anticipated audit types. Auditability implementation should be planned early in designing and developing an AI system.

Finally, we identify a set of knowledge and skills required for conducting AI audits based on interview participants' practical experiences. As noted in Hughes (2022), the pool of talent for technology professionals with AI expertise is reportedly small. Individuals with a combination of the required AI audit related skills can be rare (CSSF, 2018). Fortunately, AI audits are team efforts. In addition to AI engineers, AI architects, data scientists, data engineers, software engineers, domain experts, and product and delivery managers, our findings call for the involvement of IT auditors, subject matter experts, business and industry experts, compliance experts, social scientists, psychologists, and legal experts to fully understand and interpret the social implications of AI. The identified knowledge set appropriate for AI audit practitioners will help shape educational practices for academia and industry. Knowing the set of knowledge and skills required for AI audits will inform AI audit leaders and business clients, help them recruit the right people on the audit team, and promote audit quality. Universities should also utilize the knowledge and skills necessary for AI audits to establish and update curricula and train future generations of AI auditors.

6. CONCLUSION

This study explores measures that contribute to effective and efficient AI audits based on 23 semi-structured interviews with AI audit experts. In this study, we propose an AI auditability framework that guide the implementation of auditability measures for AI systems to ensure AI ethics, lawfulness, and robustness. We verified proposed auditability measures in the literature with experienced AI audit professionals. Compared to IT auditability measures, AI auditability demands more comprehensive information about the nature, process, quality assurance, and

governance of training data, detailed process and governance information about AI model commissioning, development, deployment, and long-term monitoring, and the governance structure relevant to developing and managing the AI system. We also added to the literature by adding novel auditability measures suggested by field professionals, such as explainability measures, mappings of causal relationships, documented model selection, deployment, long-term monitoring, governance structures, data accessibility and quality measures, and sufficient and reasonable log practices. We also identify a comprehensive set of knowledge and skills required for conducting AI audits that can guide AI audit leaders and universities to recruit and train AI audit professionals.

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8. APPENDICES

Appendix A: Interview Protocol

Table A1: Interview Protocol

No.	Interview Questions
1	At which firm are you currently working?
2	What is your role at the firm?
3	Please share with us your experiences related to AI assurance and audits.
4	What do you mean by an AI audit?
5	Please describe the process for a recent AI audit engagement.
6	Do audit procedures vary among different types of AI systems?
7	Who are the key stakeholders that auditors should communicate with during an AI audit engagement?
8	Who is expected to conduct and be involved in an AI audit?
9	What are the key knowledge and skills required for successful AI audits at each audit stage?
10	Please describe the important measures that make the AI system auditable.

Appendix B: List of Qualitative Evidence

Table B1: Qualitative Evidence for AI Auditability

No.	Auditability Characteristics	Freq.	Description	Quota
1	Auditing Standards	12	There are well-stated standards (external or internal) that informs the audit objectives and process	• "... IIA Artificial intelligence framework ..." (2320801) • "There is GDPR for data security." (2320806) • "...US FDA Audit for medical services..." (2320809) • "European AI Acts, which is a legislation by European commission, that will be applicable to European countries on how to use AI systems based on a risk-based approach. It will streamline future audits." (2320811) • "There really aren't external standards to follow." (2320814) • "NIST is working on developing a framework, that's one step in the right direction." (2320815) • "The Cross Industry Standard Process for data mining. It's about data mining, but it's the checklist. So things like understanding the business needs, understanding the data, preparing the data, modeling it, evaluating it, and of course, deploying it as it goes along." (2320818) • "Because now there are in place regulations such as GDPR, which is relevant for data protection. There are human protection laws that could be applied in the context of AI." (2320819) • "I don't think that there is any to my knowledge. I haven't found any regulatory guidelines. So each firm, I think follows its own guidelines, they develop their own. So our firm has certain guidelines. So I think that's like totally dependent on firms and what they find it acceptable actually." (2320820) • "...mostly, we followed the GDRP regulations for that data." (2320821) • "...New York Act around hiring software..." (2320822) • "Financial, healthcare industrys' own regulations." (2320823)
2	Documentation	14	Auditability is enabled by sufficient documentation	• "Schedule meetings with data manager, client manager, and AI developers to gather and document information of ..." (2320802) • "Documentation is super important. If there is no good documentation to an AI system, it is going to be very hard, and it's going to take a lot of time for the audit team to understand what the system is intended to do." (2320804) • "Basically, for each and every step we take, it needs to be well documented." (2320805) • "... all of these need to exist and be documented." (2320810) • "The better documented this process is, the more transparent the system is, the easier for us to build our test cases, and with good audit quality of it." (2320811) • "It got to be documented." (2320812) • "You need to have detailed documentation, you need to have detailed maps to..." (2320813) • "To understand the intended use, a lot of the supportive information lives in like Powerpoints." (2320814) • "And surprisingly always they don't have adequate documentation." (2320815) • "... and documentation is often the same issue where documentation is often light for the same issues as the audit is light..." (2320816) • "... and there would be regular testing of the model within that. So I would be looking for the presence of those kinds of artifacts ... and if those things aren't in place and documented, then it's very difficult to justify that there's good governance structure in place." (2320817)

				• “So these things, if it's documented well enough, auditing gets much easier because you have something to audit against.” (2320818) • “... you move into documenting data and model.” (2320819) • “When it comes to AI, having documentation is incredibly important. If the company has no documentation on how the system was built. It takes a really long time to audit. Because you basically have to go back in history and try and figure it out and piece it together. You gotta have the documentation.” (2320822)
3	Logs	6	Maintaining appropriate logs enables auditability	• “And in the end, it's traceability, replicability.” (2320823) • “The first thing is like the equivalent of logging everything that you've done. And so one of the big parts is data ... You just say all the data is in the database, what you want with it and build a model with it. I think that's probably the most important part - having a trail of this is how I get X data point; this is how it's fitted into my model; and this is how I predict. And every step in there is very specific. So it's like database ingestion logs, transformation logs to be put into your database. When was the data access? All these things are very important.” (2320814) • “We also need to have the statistics of the data. One should be mentioned here is the log files, what's happening, what happened, and who has monitored, are the right people are using the models, whether they have the right identification and registration for that particular AI models, and the data has not been used outside of the scope.” (2320806) • “...one should be mentioned here is the log files, what's happening, what happened, and who has monitored, are the right people are using the models, whether they have the right identification and registration for that particular AI models, and the data has not been used outside of the scope...” (2320806) • “Logging each and every update that we are making, and why we are making those updates, and particular disciplinary standards in order to apply the codes as well is also important.” (2320805) • “The logging depends on what is your model supposed to do, it is not necessary that every decision has to be logged. The log data could be too much and not each data is super useful, because you have to do a whole set of post-processing afterwards. But on the other hand, if you have no logging information, the audit also becomes impossible, because you don't get any metrics outside of the outputs. You need to have appropriate loggings, that means if you have high quantity model, you make sure everything that needs to be monitored is logged. You will choose a logging model that is appropriate for the model and for future monitoring and auditing, it catches the performance metrics that are necessary, for anything that is strange and adverse. The audit efforts are put into anomalies that has been detected, which is a reasonable number of data to look into but also result in statistical significance.” (2320811)
4	Model Intended Use	7	Auditability is enabled by the documentation of intended uses for AI	• “... but the intention, in my opinion, matters as much as the execution.” (2320804) • “So I think it's essential to understand, for example, something that I see missing a lot is even clearly defining what is the scope of this AI system, in which the domain companies are planning to apply, and what is the exact intended purpose of this system. Because I think many issues are emerging when it gets deployed in the wrong context and produces unexpected consequences. So I think for me, audit starts very early on into defining clearly what the trajectory companies are planning for your system.” (2320819) • “The intended use of it is very essential, especially for highly sensitive data.” (2320805) • “Understanding what the business use case was, for what organization wanted to create this model, allows you to audit it. Sometimes managers get in the way and say, I'm looking for this specific result, I am looking to optimize my trial by X amount. And doing that, you can begin to understand why we were causing biases; what were the incentives that were making these models in a certain way.” (2320814)

				• “... it's not just the AI, but how the AI is being used, who is going to use it, how often it is being used, for what purposes it is being used, all of that becomes critical.” (2320810) • “The better your model is described in how it is being created, how is the long-term use, the easier for auditability” (2320811) • “If you understand the intention of the design, whether with very detailed documentations, or the knowledge transfer meetings, if you have sort of detailed pre-context on what it is trying to do, and the reason of the project, this kind of background context really helps you to evaluate the algorithm.” (2320808)
5	Model Explainability and Transparency	7	Explainability and transparency enable auditability	• “The first thing would definitely be explainability, especially kind of the direction the field is going ... So definitely explainable would be the first kind of big factor of making something more relatable.” (2320807) • “I see no explainability, which is to me a big problem. I need to know how this model does what it does. Like a black box is not acceptable.” (2320815) • “No one really understands the criteria neural networks actually use for their predictions ... Of course there are specific ways you can get some insight through kind of training, but at the same time those algorithms are not strictly explainable.” (2320807) • “Whereas things like neural networks that's not really the case. So, in terms of transparency, there is no transparency in that case, so you lose that, you have to find other means of trying to validate that it does what it says on the box.” (2320817) • “Commented sections about each step that AI is doing as much as possible so we can interpret. It is not necessary that we'll be able to interpret each and everything out of it. Sometimes it is like a black box, but other times it's just plain and simple. We can reduce our model in such a way that it is auditable, and it's simple to understand by anyone.” (2320805) • “... the auditable AI dashboard should be designed to ensure that model explainability or the overall explainable AI.” (2320823) • “So in those cases, it's a little bit of a brute force approach and there are many, many methods and approaches to do it. And one brute force method that at least I used has been to build a parallel ensemble machine learning model with all of the features ... so that we at least know what features are going in, and how they may typically load, as far as the model outcomes might be concerned.” (2320823)
6	Model Selection and Model Values	4	Auditability is enabled by the documentation of model selection process and model values	• “Is this completely automated? Or is this something that is only augmenting human decision making? So, to what extent human is in the loop or is over the loop. All of these need to exist and be documented.” (2320810) • “...a very important part is, the model selections parameter, which you're using.” (2320821) • “Whatever model they use, we can't go into the events and find out what exactly it is doing with the data. But we have to set particular standards as to why we are using it to back up each decision of ours. We are to test different methods in order to say that, yes, this is the particular approach and this is why we are talking this approach. This would make it very simple for the audit.” (2320805) • “... what are the contributions... what readily available market products which can replace their existing models; whether the model could be beneficial and whether they can have a third-party model rather than building their own custom models.” (2320802)
7	Model Operations	8	Auditability is enabled by the documentation of model operations and historical	• “... to gather and document information of ... what are the models they have trained with ... what are the results” (2320802) • “Like the whole overall processes including the data and the model they are using, how they use them.” (2320805)

			performance results	• “What is their model? How was the model performing? And not just when it was built, but also on an ongoing basis, to make sure it doesn't deteriorate in quality over time.” (2320815) • “... you need to have detailed maps to show all the causal relationships, that happen even before you get to build the AI system ... being as detailed as possible ... we are basically looking at all the different flows, and how all the causal relationships go together.” (2320813) • “Another growing topic that one never does is the causality. That's a big topic as well. That's data scientists tend to forget about. Causality is very important. I mean, like, if you see some kind of relationship between an input and an output factor. Is it the X that drives the Y, or is it not.” (2320815) • “... what is being produced at each stage of that process at the business side, at the data side, building the model, deploying the model, what are the different artifacts of the deliverable needs to be produced.” (2320810) • “What the algorithm has historically done would be important as well, because there's always some type of random noise when you're running an algorithm.” (2320804) • “Does the organization have a culture of risk management or not, because if it does, we expect to see things like risk forums; and depending on which, someone will have done a risk assessment on the model and the product typically that it's embedded within, and have a set of again defined thresholds for what it looks like from a performance point of view, and there would be regular testing of the model within that. So I would be looking for the presence of those kinds of artifacts. Is there testing being done? What are the results saying? What is management doing based on the results that are coming back? How are they identifying and addressing any issues that are being identified. And if those things aren't in place and documented, then it's very difficult to justify that there's good governance structure in place.” (2320817)
8	Model Deployment and Monitoring	2	Auditability is enabled by the documentation of model deployment and monitoring	• “... what is being produced at each stage of that process at the business side, at the data side, building the model, deploying the model, what are the different artifacts of the deliverable needs to be produced...” (2320810) • “Another important aspect is to provide details on how the model is embedded into the overall product and process. It is not only the model performance itself, but also how the model is monitored, how do you measure the performance of the lifetime of the product, and what kind of measures have you foreseen in terms of re-training the model and figuring out when your model fails.” (2320811)
9	Data Accessibility	4	Auditability is enabled by providing training data	• “The most important thing with AI in general is they forget about importance of clean data; you have to be able to know the data it's working with. The hardest part for us to implement AI audit is being able to have the data.” (2320812) • “It means you are transparent on what kind of training data you have used, it doesn't mean you have to provide us all the training data, but it has to be well described ... We need some samples of the data in order to see if there is something obviously adverse in the training data.” (2320811) • It has to be easy enough to pull out the needed data to save auditors' time and efforts. For instance, an interview (2320816) stated, • “The ease at which you can pull data like a lot of systems will have their historical data and their forecast data or whatever it is, or their predictions are in totally different tables.”
10	Data Characteristics	2	Auditability is enabled by the documentation of data characteristics	• “I think one of the most important parts is the characteristics of the data.” (2320821) • “We all know the training data itself and the AI model can be extremely extensive, therefore it has to be described in a way that all the data features and uses have been captured.” (2320811)

11	Data Process and Governance	7	Auditability is enabled by the documentation of process and governance surrounding data	• “... to gather and document information of how did they come to data.” (2320802) • “Like the whole overall processes including the data and the model they are using, how they use them.” (2320805) • “It is also about the transparency of the processes in which the AI model has been trained, and providing the typical forms of metrics that has been generated, essentially highlighting what steps you have taken to train this model, and how well it works. So that people can confirm the performance by doing our own testing following the guide that has been provided.” (2320811) • “How was data sourced and cleaned? How was it used?” (2320815) • “... make sure that there is transparency on how the training data was being put together ... How did we come up with the model? Were all these things taken into account: privacy, transparency, accountability? Who's taking ownership? Are these documented? The way the training model was perceived? Was there the right quality of data?” (2320818) • “The data governance should be very robust, which means the data quality, data security, data trail, data compliance should be adhering to the whole set of policies and security aspects. This will make the audit process very more systematic.” (2320806) • “... what is essential is to even find quality measures, so develop methodologies that can be implemented to actually assess the quality of the data you're feeding to the model, because currently there is also like a lot of discrepancy. So, for example, some companies buy data from an external provider, but they don't have any methodology to actually assess the quality of this data that they're feeding to the model developed.” (2320819)
12	Clients' Personnel Competency	2	Clients with a competent developing team and other relevant personnel to design, develop, deploy the AI system enhance auditability	• “Developers and project managers should know, first, what are security constraints for where they are deploying; second, the process they have used for; third, the frequency of the model being used; fourth, how many times are they invoking the model; fifth, how much data can be handled.” (2320802) • “Dealing with technical debt, where you could see something happening and you could write a fix in Python. But then actually making it fit in with the code base was then 80% of your time because the flexibility of the code base was so limited and then same with the debug and then you would spend a lot of time just debugging. What they had was a software language that was so much more complicated and so less flexible, you'd be better off asking your mathematicians to learn whatever software language. So technical debt is an issue that cuts into audit time too.” (2320816)
13	Clients' Governance Structure	5	Auditability is enabled by the documentation of governance structure	• “Is there a good governance standard? So this kind of governance, and is that being updated? I think, on top of that, what is your continuous assurance and continuous compliance model? ... If it's documented well enough, auditing gets much easier because you have something to audit against.” (2320818) • “It's just fundamentally not described, like the governors is not described. Who's responsible? Who does what? ... And what's your governance around it? What are your thresholds, your metrics for what is acceptable performance.” (2320815) • “Who is authorizing all those deliverables, and how does that authorization get escalated from data scientist to leader and then CEO, CXO? What is the escalation process, and what are the controls? So, all of that is what makes a program successful, not just having a few people coding.” (2320810) • “And how the management structures of the organization using what comes out of that to make decisions? Is there a certain point with which the organization is set if the confidence of the machine is below a certain threshold then it falls out to a human decision maker, or is there a periodic review as well of decisions that have gone through?” (2320817)

				• “We would also want to be looking at not just the model itself, but all the governance structures around it. So how is it being fit? What happens with the information that comes out the other side of it? How is it being looked after and governed and managed and restricted appropriately.” (2320817)
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Table B3: Qualitative Evidence for AI Audit Knowledge and Skills

No.	Knowledge and Skills Captured	Freq.	Description	Quota
1	Knowledge of compliance requirements	13	AI auditors should be knowledgeable about the compliance requirements for AI	• “Subject matter expertise, industry expertise, industry standards, and some common issues associated with this AI system.” (2320804) • “We are to make sure that none of the regulatory standards are compromised, so we need to have a knowledge of that side as well.” (2320805) • “Governance and accountability have to be there. The auditor should also know the data production rules.” (2320806) • “You would definitely need someone in terms of privacy in a team to make sure all of those information in the data is being safely stored.” (2320807) • “... what are the some of these policy documents saying? For example, there is a lot around critical recognition what is allowed what is not allowed. So with the EU and some other regulators, they need to be aware of some of those.” (2320810) • “Also compliance, the requirements kind of thing. You are being able to certify that a model or a system and AI capability is trustworthy, and it's not injecting its own bias. And the compliance requirements are industry-specific, it's very difficult.” (2320812) • “So then you have the third pillar is like a regulation and frameworks. And also understand the frameworks that you use.” (2320815) • “Is there any regulations or things that are necessary to be mindful of in terms of how this thing is looked after and governed and managed?” (2320817) • “You should know internal compliance functions that assure bias on algorithms and impact.” (2320818) • “...find out different framework...” (2320820) • “And then I would say, the last one is probably someone with a background in policy, compliance, regulation. That's not as important right now, but it's going to be very important in about a year.” (2320822) • “On the other side, the auditors, there are generally going to be more lawyers and coming from legal and risk background, they will have to be coached and trained in understanding explainable AI, at least the basics and the fundamentals of it too, so that they can at least understand what is being presented to them because there is a significant knowledge gap that exists.” (2320823)
2	AI knowledge	11	AI auditors should have AI knowledge	• “It's important for all team members to know basic theoretical knowledge, be able to explain what artificial intelligence does.” (2320801) • “...what is AI ...” (2320802) • “... skilled in ... AI and machine learning...” (2320804) • “The first thing is to have AI knowledge. Like what the AI and ML is, how the model is performing, and what's the problem statement, what's being predicted. The auditor whoever is performing the AI audit should have the domain knowledge.” (2320806) • “...skills they should definitely have is the knowledge on AI.” (2320809) • “... you need the technical expert who knows how the AI works, and knows what kind of issues can come there. When it comes to assessing the AI model, you really need a quite deep expertise on how AI works in general.” (2320811)

				• "... someone who's knowledgeable in the field of statistics and machine learning to understand what are the causes that will cause a model to be biased." (2320814) • "It's somebody who knows statistics and machine learning and somebody who knows enough about coding to understand how to write the tests and how to execute them." (2320816) • "You obviously need to be familiar with what the different types of technologies are. So that's a really big first part of it." (2320817) • "Computer Science; understanding how these technologies operate..." (2320819) • "If it is the AI engineering team, they need to be well versed in evolving methodologies and explainable AI, and then being able to code and implement and create dashboards. So that's the technical and mathematical skills." (2320823)
3	Knowledge of the underlying AI model being audited	9	AI auditors should be knowledgeable about the underlying model being audited	• "Basic understanding of AI audit project and the process, first, technical details about the functional part of the ML, at least what type of the model it is, how it works, what type of dataset was used; second, risk of the model - what is the impact that the company faces and how to evaluate and calculate the impact; third, impact minimization and mitigation; fourth, what type of information is being shared." (2320802) • "At least a basic understanding of what that AI model is doing, and why it is doing something, to make sure it is going correctly." (2320805) • "...first of all, the kind of the technology that's being used." (2320807) • "You should understand where it is applied, how it is applied, how were you doing it, which sector it is doing it. And you have to have the knowledge of the industry to analyze the impacts of AI." (2320809) • "They need to understand how some of these models work so that they can start asking the right questions." (2320810) • "You need to be technical enough to be able to understand what you're looking at." (2320812) • "And understand how those systems work. So you can't audit them if you don't understand how they go together." (2320817) • "...to understand these models." (2320820) • "...knowledge related to the metrics of the models ... understanding the variables." (2320821) • "you need to have someone that specializes in that specific system to be able to audit it, or at least has specific experience in that. It's hard to audit a system that you have to learn at the same time of what it is." (2320822)
4	Business knowledge	9	AI auditors should be knowledgeable about the client's business process	• "...basic business knowledge." (2320802) • "I believe the understanding the current state of business operations and the landscape of their data analytics platform." (2320808) • "One is they need to understand the business as to what the business purposes, how the models are being used, what are the users, how they are using the model. I think that's critical. So the business knowledge plus the data science knowledge would be the key." (2320810) • "You also need a business expert who knows how the overall business process works in order to successfully execute that." (2320811) • "...and then four is the business domain, the business expertise." (2320815) • "People who have good business knowledge ..." (2320816) • "So there's to some extent important knowledge around what the organization is, the use case of the product is, what industry do they work in." (2320817) • "...understanding the business..." (2320821) • "The other thing is what I've seen with experience is that at least on the business side..." (2320823)

5	Data science expertise	8	AI auditors should have expertise in data science, statistics, and mathematics	• “Data science and ... basic theoretical knowledge...” (2320801) • “Skilled in data science...” (2320804) • “We are doing an analysis of, let's say, k-means clustering, or a decision tree, or deep learning model. They need to understand how some of these models work so that they can start asking the right questions. So the business knowledge plus the data science knowledge would be the key.” (2320810) • “The profile would be data scientists that are very technical...” (2320811) • “The other one is someone who's knowledgeable in the field of statistics and...” (2320814) • “... somebody who knows statistics or whatever...” (2320816) • “...data modeling, statistical experience...” (2320618) • “... the technical and mathematical skills.” (2320823)
6	Data engineering expertise	6	AI auditors should have data engineering expertise, including database and systems, data modeling, data sampling, and data analytics	• “...database and systems.” (2320803) • “... data analytics and software skills” (2320806) • “I think you need someone who understands first how the data was gathered ... where does that process go? And then what do we do to get the data into our database? That's the first skill you need. I'll call like that investigator or data domain expert.” (2320814) • “...understanding of working with data.” (2320815) • “... you need to be fairly analytical and kind of in tune with what's going on in order to make those decisions, and to understand what the nature of the deviations potentially are; what those implications are.” (2320817) • “... understanding the data; understanding that data leakage point, that's very important for you.” (2320821)
7	Knowledge of social implications	5	AI auditors should have a good understanding of social implications of AI technologies	• “And finally you would need definitely a sort of social scientist person depending on the obligation. That person was responsible to identify biases against specific groups, minorities, and other things.” (2320807) • “... It's someone who understands the outcomes of this, and can understand what would actually happen if we predicted a certain way or not. If there's bias within our model, and we're not auditing our models correctly, what are the risks, the outcome expert...” (2320814) • “I would also say socioeconomic experience.” (2320818) • “What potential ethical consequences can emerge; how can they develop methodologies to address these impacts, to mitigate them early on; you need to translate some ethical requirements or legal requirements in the case of the AI act that are coming into mathematical formulations into code ... There should be psychologists. I see a role emerging because we are interacting more and more, we are exposing children to these technologies, so really preventing harmful content, these are big challenges that need to be addressed.” (2320819) • “We're keeping the high-level perspective coming in, which is the direction, the vision of the system.” (2320822)
8	Auditing skills	5	AI auditors should understand the auditing process and have auditing skills for AI audit projects	• “... what is the requirement of AI audit ... basic ML standards and evaluation metrics: how it is calculated, how it is performed...” (2320802) • “... knowing how to measure and assess AI systems ... traditional sense of auditing...” (2320804) • “The primary set of skills would be auditing.” (2320806) • “And the fifth is the mindset of validating and auditing.” (2320815) • “... and then the very standard consulting type things like stakeholder management, communication, writing up a report.” (2320817)
9	Coding competency	4	AI auditors should have	• “...programming skills ...” (2320815) • “... and somebody who knows enough about coding to understand how to write the tests and how to execute them.” (2320816)

			strong coding competency	
10	Competency in AI audit tools	2	AI auditors should know how to use audit tools in AI audits	“... to have the necessary tools in order to make sure that you're capturing all the information ... Have the necessary tools to perform all the audit procedures ...” (2320813) “Right now the current people can't directly look into this model without a good framework and providing them tools.” (2320820)
11	Software engineering skills	2	AI auditors should have experiences in software engineering	“... and software skills ...” (2320806) “So combining everything in terms of like knowing... knowing software engineering...” (2320807)
12	Soft skills	2	AI auditors should have social goodness and storytelling skills	“The passion, the motivation, the whole thing should be put into any AI auditing. The whole thing is social goodness. The first and primary concern of anybody who does it is whether that person is socially good or not. That is the first thing. The second thing is the faithfulness, are we faithful to people who concern. This is more than any technical expertise. Technical expertise, software skills come later. The first thing of the audit person is being clean, being pure, definitely we need a good auditor.” (2320809) “... then a skill will be required in storytelling. Being able to take those complex statistical outcomes or mathematical outcomes and being able to translate those into a business language that can be understood, that is where a lot of the failures also happen...” (2320823)